

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 7. String and its Methods

Lecture: 80. Restaurant Menu - Project

Section 1: Lecture Summary

This lecture introduces a practical Python project centered around creating a restaurant menu display. Unlike previous small problems termed as challenges, this project requires utilizing basic Python features to produce a meaningful, formatted output. The task is to write a Python program that takes four menu items along with their prices as input, and prints them in a standardized menu format. The key requirement is to display each item name and price on the same line separated by dashes, where the total length of each line is fixed (e.g., 20 characters). The number of dashes is calculated dynamically based on the length of each item and price. The instructor explains the logic for calculating the number of dashes and how to print the formatted string by concatenating the item name, a string of dashes (using string repetition), and the price. A partial example showing these calculations and print statement is presented, and students are encouraged to complete the program for all four items.

Section 2: Key Concepts and Explanations

- Project Objective

- Write a Python program to print a restaurant menu of four items and their prices.
- Each line displays the item name, followed by dashes, then the price.
- The total length of each printed line is constant (e.g., 20 characters).

- Input Handling

- Four item names and their prices must be taken as input from the user.
- Both item names and prices are treated as strings.

- Fixed Width Formatting with Dashes

- The total line length is fixed (e.g., 20).
- Calculate the number of dashes by subtracting the lengths of the item name and price strings from the total length.
- Use string repetition (`'-' * number_of_dashes`) to create the dash segment.

- String Concatenation

- Combine the item name + dash string + price into a single string for printing.
- Concatenation is done using the `+` operator.

- Length Calculation

- Use the `len()` function to find the number of characters in the item name and price strings.

- Example Calculation

- For item name "Hot Dog" (7 characters) and price "30" (2 characters),
- Number of dashes = $20 - 7 - 2 = 11$ (example in transcript said dashes as 10 but exact depends on exact counting).
- Final string would be `"Hot Dog" + "-----" + "30"`.

Section 3: Example Code and Use Cases

Example for one item

Fixed total length for each printed line

Calculate number of dashes needed

Create the dash string by repeating '-'

Concatenate the item name, dashes and price and print

```
item1 = "Hot Dog"
price1 = "30"

total_length = 20

dash_count = total_length - len(item1) - len(price1)

dashes = '-' * dash_count

print(item1 + dashes + price1)
```

Explanation:

This code snippet prints one menu item formatted with dashes separating the item name and price to maintain a total length of 20 characters.

Full program template for four items

Taking inputs from user

Print all four items

```
item1 = input("Enter item 1 name: ")
price1 = input("Enter item 1 price: ")
item2 = input("Enter item 2 name: ")
price2 = input("Enter item 2 price: ")
item3 = input("Enter item 3 name: ")
price3 = input("Enter item 3 price: ")
item4 = input("Enter item 4 name: ")
price4 = input("Enter item 4 price: ")

total_length = 20

def print_menu_item(item, price):
    dash_count = total_length - len(item) - len(price)
```

```
dashes = '-' * dash_count
print(item + dashes + price)

print_menu_item(item1, price1)
print_menu_item(item2, price2)
print_menu_item(item3, price3)
print_menu_item(item4, price4)
```

Explanation:

This complete sample program takes input for four items and their prices, then prints each formatted line using a function to reuse the dash-calculation logic.

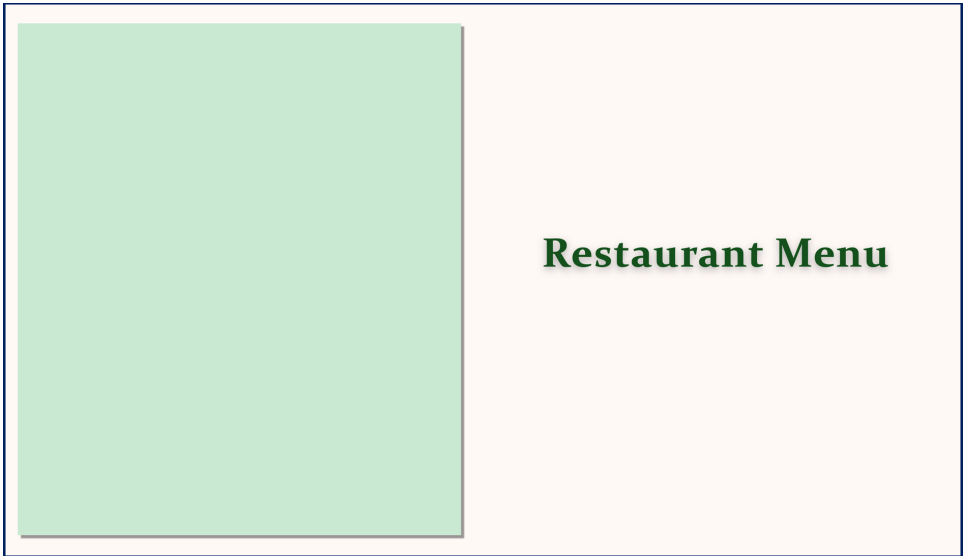
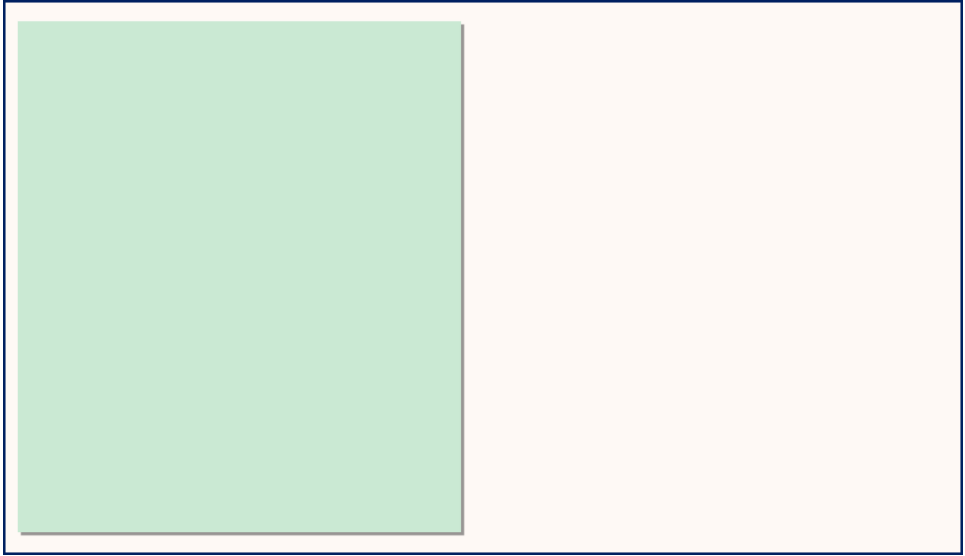
Section 4: Key Takeaways

- This project emphasizes formatting output with precise alignment using string length calculation and string repetition in Python.
- Fixed width formatting can be achieved by calculating and filling gaps with characters (in this case, dashes).
- String concatenation (`+`) and multiplication (`\*`) operators are effective tools for building such formatted strings.
- The `len()` function is essential for dynamic formatting based on input length.
- Taking input as strings avoids unnecessary type conversions since prices are printed as strings, not used in arithmetic here.
- The project introduces the concept of creating small reusable functions to handle repetitive tasks.
- This task bridges fundamental string operations with practical output formatting useful in real applications.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



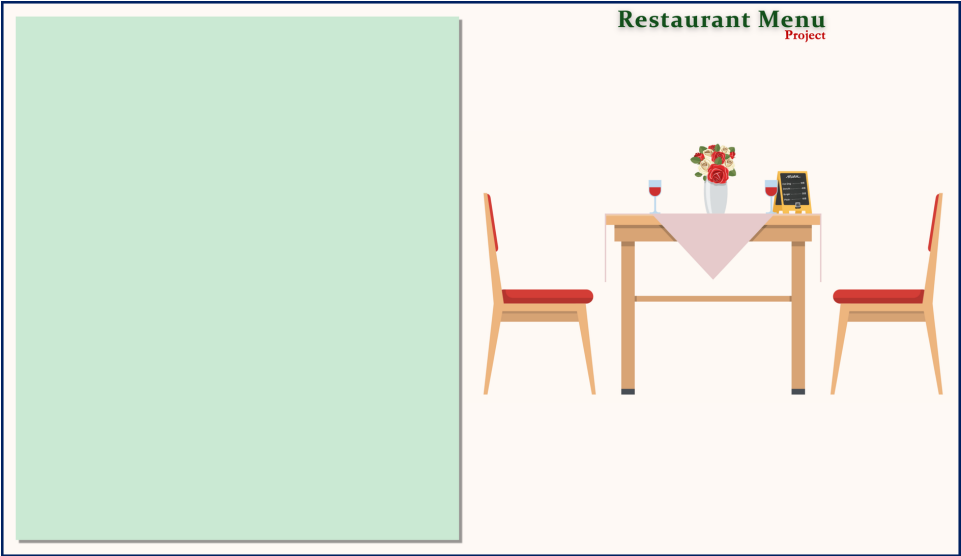


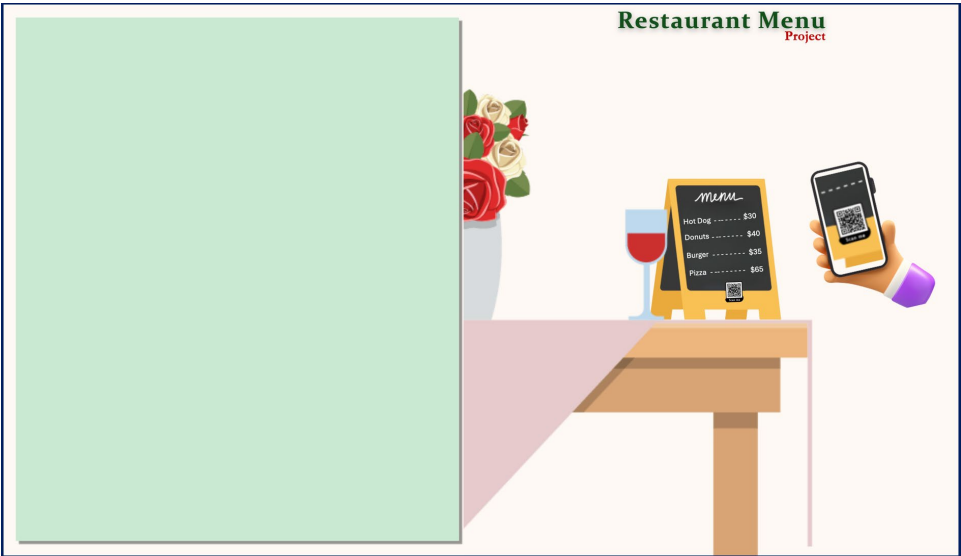
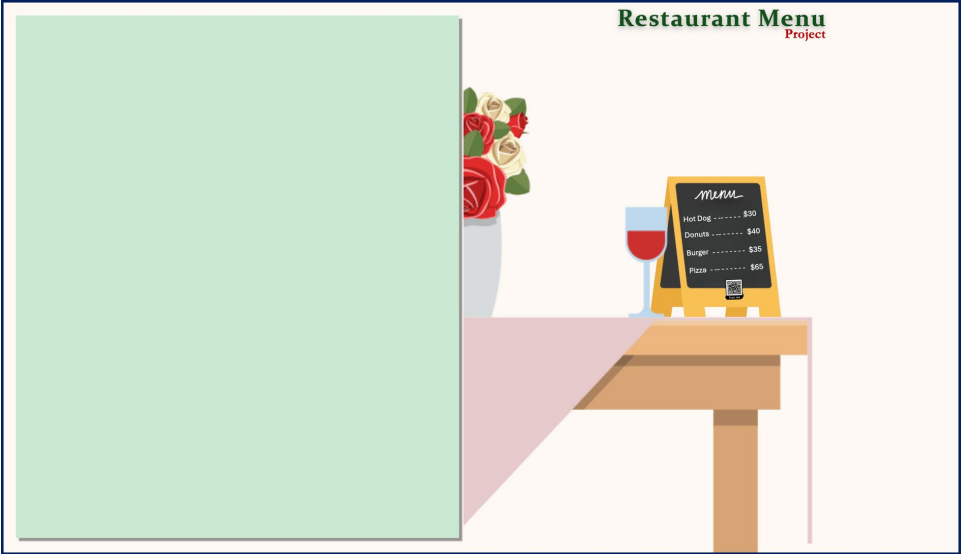
Restaurant Menu

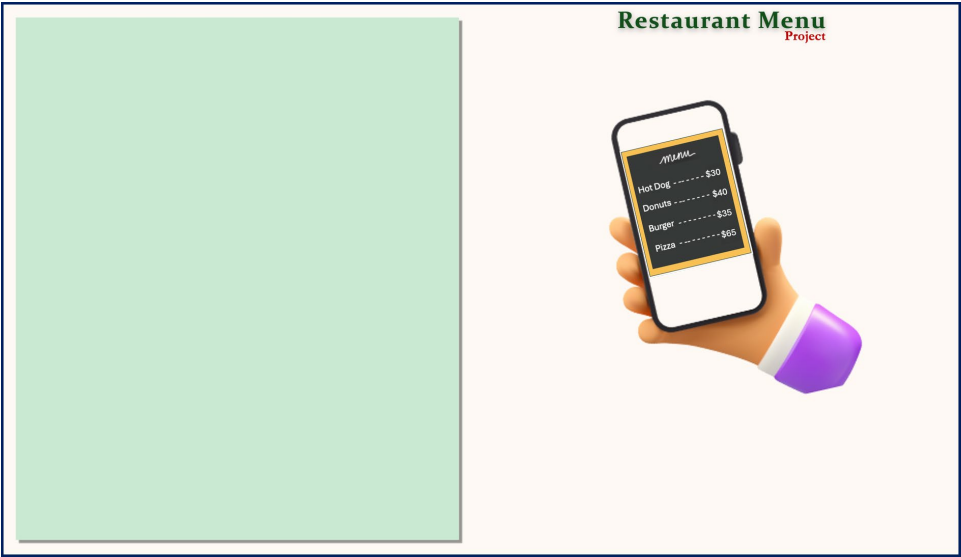
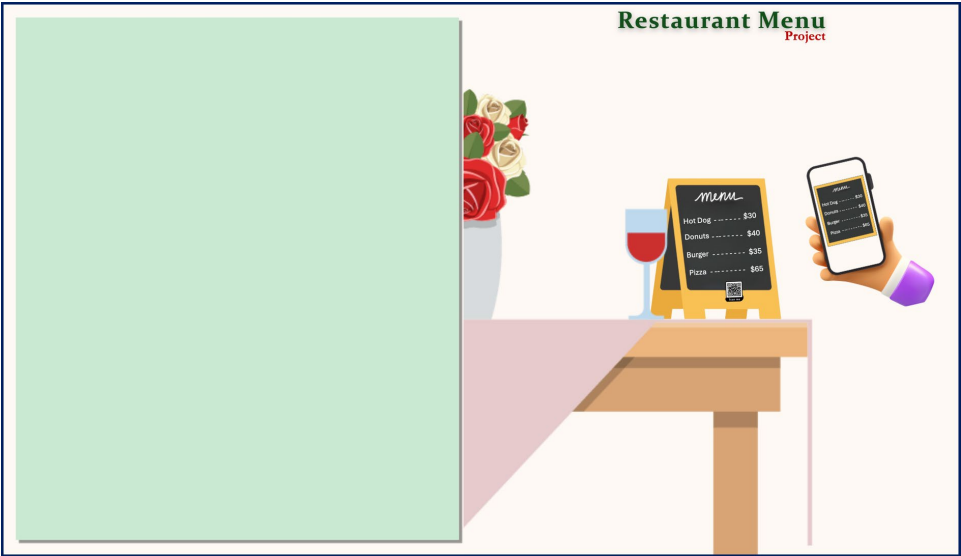
Project

Restaurant Menu

Project









Restaurant Menu

Project

Hot Dog \$20

Donuts \$40

Burger \$35

Pizza \$65

Restaurant Menu

Project

Hot Dog \$20

Donuts \$40

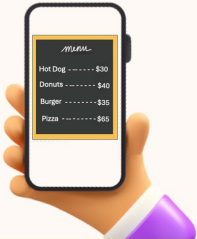
Burger \$35

Pizza \$65

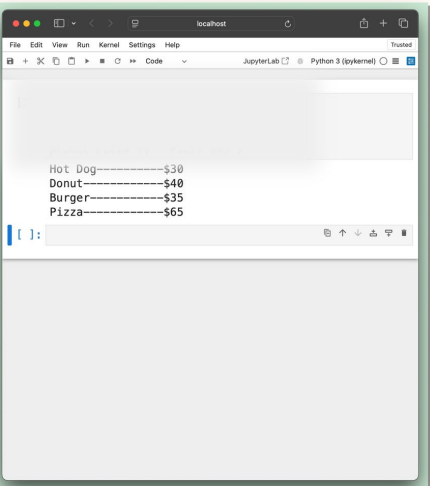
item

Restaurant Menu

Project



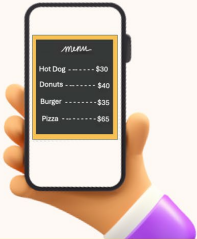
item ----- price



```
Hot Dog-----$30
Donut-----$40
Burger-----$35
Pizza-----$65
```

Restaurant Menu

Project

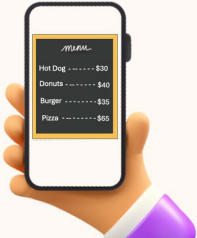


Hot Dog ----- \$30

7 ← 10 → 3

Restaurant Menu

Project



7

Hot Dog

10

\$30

3

len(item1)

dash =

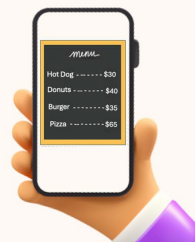
len(price1)

Restaurant Menu Project



7 10 3
Hot Dog ----- \$30
len(item1) len(price1)
dash = 20

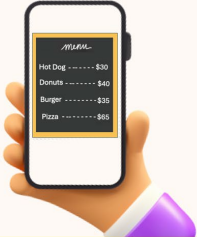
Restaurant Menu Project



7 10 3
Hot Dog ----- \$30
dash = 20 - len(item1) - len(price1)
' - '

Restaurant Menu

Project



7 ← 10 → 3

Hot Dog ----- \$30

$\text{dash} = 20 - \text{len}(\text{item1}) - \text{len}(\text{price1})$

$\text{'_'} * \text{dash}$

Restaurant Menu

Project

7 ← 10 → 3

Hot Dog ----- \$30

$\text{dash} = 20 - \text{len}(\text{item1}) - \text{len}(\text{price1})$

$\text{'_'} * \text{dash}$

Restaurant Menu Project

7 ← 10 → 3
Hot Dog ----- \$30

dash = 20 - len(item1) - len(price1)

'_' * dash

item1 = 'Hot Dog'

Restaurant Menu

Project

7 ← 10 → 3

Hot Dog ----- \$30

$\text{dash} = 20 - \text{len}(\text{item1}) - \text{len}(\text{price1})$

$'_ '* \text{dash}$

item1 = 'Hot Dog'

Restaurant Menu

Project

7 ← 10 → 3

Hot Dog ----- \$30

$\text{dash} = 20 - \text{len}(\text{item1}) - \text{len}(\text{price1})$

$'_ '* \text{dash}$

item1 = 'Hot Dog'

price1 = '30'

Restaurant Menu
Project

7 ← 10 → 3

Hot Dog-----\$30

dash = 20 - len(item1) - len(price1)

'_' * dash

item1 = 'Hot Dog'

price1 = '30'

Restaurant Menu
Project

7 ← 10 → 3

Hot Dog-----\$30

dash = 20 - len(item1) - len(price1)

'_' * dash

item1 = 'Hot Dog'

price1 = '30'

dash

Restaurant Menu
Project

7 ← 10 → 3

Hot Dog-----\$30

dash = 20 - len(item1) - len(price1)

'-' * dash

item1 = 'Hot Dog'

price1 = '30'

dash = 20

Restaurant Menu
Project

7 ← 10 → 3

Hot Dog ----- \$30

$\text{dash} = 20 - \text{len}(\text{item1}) - \text{len}(\text{price1})$

'-' * dash

item1 = 'Hot Dog'

price1 = '30'

$\text{dash} = 20 - \text{len}(\text{item1})$

Restaurant Menu
Project

7 ← 10 → 3

Hot Dog ----- \$30

$\text{dash} = 20 - \text{len}(\text{item1}) - \text{len}(\text{price1})$

'-' * dash

item1 = 'Hot Dog'

price1 = '30'

$\text{dash} = 20 - \text{len}(\text{item1}) - \text{len}(\text{price1})$

Project

```
dash = 20 - len(item1) - len(price1)

'_' * dash
```

```
dash = 20 - len( item1 ) - len( price1 )
```

Project

```
dash = 20 - len(item1) - len(price1)

    ' _ ' * dash
```

```
print( )
```

Project

' _ ' * dash

```
print( item1 )
```

Restaurant Menu Project

7 ← 10 → 3
Hot Dog ----- \$30

dash = 20 - len(item1) - len(price1)

'-' * dash

item1 = 'Hot Dog'

price1 = '30'

dash = 20 - len(item1) - len(price1)

print(item1 + '-' * dash)

Restaurant Menu Project

7 ← 10 → 3
Hot Dog ----- \$30

dash = 20 - len(item1) - len(price1)

'-' * dash

item1 = 'Hot Dog'

price1 = '30'

dash = 20 - len(item1) - len(price1)

print(item1 + ('-' * dash))

Restaurant Menu Project

7 ← 10 → 3
Hot Dog ----- \$30

`dash = 20 - len(item1) - len(price1)`

`'-' * dash`

`item1 = 'Hot Dog'`

`price1 = '30'`

`dash = 20 - len(item1) - len(price1)`

`print(item1 + ('-' * dash) + price1)`